Jinwon Kim

Robotics Engineer

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Summary

- Part-time PhD researcher (company-sponsored, Korea University) on VLM-based robot autonomy and RL for legged locomotion — building sim-to-real pipelines on FuRO with MuJoCo, mjlab, and CMA-ES-based parameter identification (PACE).
- Shipped end-to-end autonomy stack on Vision 60 (SLAM/GLIM, DLIO localization, Nav2+MPPI planning, Behavior Tree missions) as Robot-Intelligence Team Lead at KRM.
- 3 government-funded R&D programs (IITP, NIPA, defense cluster), 2 corresponding-author papers, 1 filed patent, and Ministry of National Defense Drone-Bot Challenge Grand Prize (2 consecutive years, 2023 & 2024).

Experience

Korea Robot Manufacturing, Robotics Engineer (Robot-Intelligence Team Lead) — May 2023 – present, 3 years

- Led 3-5 engineers on autonomous navigation for quadruped robots (Vision 60)
- Deployed full autonomy stack on Jetson Orin / ROS 2 — GLIM (SLAM), DLIO (localization), Nav2+MPPI (planning), Behavior Tree (missions)
- Designed ROUTE — graph-based patrol mission system with plugin architecture (PTZ, docking, gait switching)
- Built PACE (Parameter Actuator Calibration Engine) for MuJoCo sim-to-real transfer. Joint dynamics error under 5% NRMSE via CMA-ES
- Contributed to 3 government-funded R&D projects (IITP, NIPA, defense cluster)

Korea Institute of Science and Technology, Student Researcher — May 2019 – Dec 2019, 8 months

- Collected 1,000+ annotated images via web crawling for delivery robot object detection
- Wrote real-time multi-object detection and tracking with YOLOv3 and Siamese network
- Published at KRoC 2020; patent registered (KR-10-2020-0026298)

Education

PhD — **Korea University**, Computer Science and Engineering — Seoul, South Korea, Sept 2025 – present

- Research: VLM-based robot autonomy, reinforcement learning for quadruped locomotion
- Part-time enrollment, sponsored by Korea Robot Manufacturing

MS — **Korea Advanced Institute of Science and Technology**, Robotics — Daejeon, South Korea, Mar 2021 – Feb 2023

Scalable Graphics, Vision, and Robotics Lab (Advisor: Prof. Sung-eui Yoon)

- Focus: Reinforcement Learning, Deep Learning, Intelligent Robotics

BS — **Kwangwoon University**, Division of Robotics — Seoul, South Korea, Mar 2017 – Feb 2021

- GPA: 4.23 / 4.5

Publications

Urban Autonomous Robot Navigation via Natural-Language Route Instructions and VLM-Based Situation Understanding — Jan 2026
Minje Kim, Kyungtae Park, ***Jinwon Kim***†
(Korea Robotics Society Annual Conference (KRoC 2026))

ATBT: Adaptive Topological Map-Based Behavior Tree for Quadruped Robots — Jan 2025
Woosung Yoon, Junhyeok Choi, Kyungtae Park, Joonghyun Shin, Seungtaek Sung, ***Jinwon Kim***†
(International Conference on Control, Automation and Systems (ICCAS 2025))

Collision-Backpropagation based Obstacle Avoidance Method for a Legged Robot Expressed as a Simplified Dynamics Model Jan 2022
Jinwon Kim, Heechan Shin, Sung-eui Yoon
(International Conference on Control, Automation and Systems (ICCAS 2022), Busan, Korea)

Patents

Robot Path Control Apparatus and Method for Determining Movement Path Considering Obstacles 2024
Jinwon Kim

Robust Multi-object Detection Apparatus and Method Using Siamese Network 2022
KangGeon Kim, **Jinwon Kim**

Projects

Sim-to-Real Pipeline for Quadruped Robot (FuRo) Korea Robot Manufacturing
MuJoCo simulation environment and PACE parameter identification for RL-based locomotion Dec 2025 – present

- PACE pipeline identifies 6 physics parameters (armature, damping, frictionloss, kp, kd, latency) from 1kHz real robot data using CMA-ES
- Set up mjlab-based RL training environment on top of PACE-calibrated dynamics for sim-to-real transfer

Quadrupedal Robot System for Defense Monitoring and Reconnaissance Agency for Defense Development (ADD)
National defense research project (KAIST MS thesis) Mar 2021 – Jan 2023

- Used deep learning to generate initial trajectories for optimization, 100x faster than prior approach

Activities

Auturbo Mar 2023 – present
Quadruped Robot Team Leader (Mar 2024 – present), Regular Member (Mar 2023 – Dec 2023)

- Building StrideSim, a quadruped robot simulation platform on IsaacSim

Honors

- Military Robotics Society Excellence Award (Dec 2025)
- Military Robotics Society Excellence Poster Award (Nov 2024)
- Ministry of National Defense Drone-Bot Challenge: Grand Prize, 2 consecutive years (2023, 2024)
- Dean's List (2020, 2019, 2018)

Skills

Programming Languages: C++, Python

Robotics & Navigation: ROS 2, Nav2, SLAM (GLIM, DLIO), Behavior Trees, MPPI

Simulation & RL: MuJoCo, IsaacLab, PyTorch, mjlab

Tools & Infrastructure: Git, Docker, Jetson Orin, Ansible, Jenkins

Language Fluency: Intermediate-high English (TOEIC 810, OPIc IH), Native Korean